

VLSM Tree:

11.79.0.0/16

|
|—— **11.79.0.0/19** ————— **Granite Borough (GB)**

| **Hosts needed: 5200**

| **Subnet: 11.79.0.0 – 11.79.31.255**

| **Usable: 8190 hosts ✓**

|
|—— **11.79.32.0/21** ————— **Corsair Isles (CI)**

| **Hosts needed: 1200**

| **Subnet: 11.79.32.0 – 11.79.39.255**

| **Usable: 2046 hosts ✓**

|
|—— **11.79.40.0/23** ————— **Frostfield Ranch (FR)**

| **Hosts needed: 350**

| **Subnet: 11.79.40.0 – 11.79.41.255**

| **Usable: 510 hosts ✓**

|
|—— **11.79.42.0/24** ————— **Aurora Ridge (AR)**

| **Hosts needed: 150**

| **Subnet: 11.79.42.0 – 11.79.42.255**

| **Usable: 254 hosts ✓**

|
|—— **11.79.43.0/25** ————— **Photon Cove (PC)**

| **Hosts needed: 80**

| **Subnet: 11.79.43.0 – 11.79.43.127**

| **Usable: 126 hosts ✓**

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|—— 11.79.43.128/26 ————— Sunspire Palace (SP)
|   Hosts needed: 45
|   Subnet: 11.79.43.128 – 11.79.43.191
|   Usable: 62 hosts ✓
|
|—— WAN Links (Serial /30s) ————— Point-to-Point Links
|
|—— 11.79.43.192/30 — R-PC ↔ R-SP
|—— 11.79.43.196/30 — R-PC ↔ R-GB
|—— 11.79.43.200/30 — R-PC ↔ R-FR
|—— 11.79.43.204/30 — R-GB ↔ R-CI
|—— 11.79.43.208/30 — R-GB ↔ R-AR
|—— 11.79.43.212/30 — R-SP ↔ R-AR

```

IP Address Table

LAN Interfaces

Device	Interface	IP Address	Subnet Mask	Network
R-PC	Gi0/0	11.79.43.1	255.255.255.128	Photon Cove LAN
R-GB	Gi0/0	11.79.0.1	255.255.224.0	Granite Borough LAN
R-SP	Gi0/0	11.79.43.129	255.255.255.192	Sunspire Palace LAN
R-FR	Gi0/0	11.79.40.1	255.255.254.0	Frostfield Ranch LAN
R-AR	Gi0/0	11.79.42.1	255.255.255.0	Aurora Ridge LAN
R-CI	Gi0/0	11.79.32.1	255.255.248.0	Corsair Isles LAN

WAN Interfaces:

WAN Interfaces

Device	Interface	IP Address	Subnet Mask	Link
R-PC	S0/0/0	11.79.43.194	255.255.255.252	PC↔SP
R-SP	S0/0/0	11.79.43.193	255.255.255.252	PC↔SP
R-PC	S0/0/1	11.79.43.197	255.255.255.252	PC↔GB
R-GB	S0/0/0	11.79.43.198	255.255.255.252	PC↔GB
R-PC	S0/1/0	11.79.43.201	255.255.255.252	PC↔FR
R-FR	S0/0/0	11.79.43.202	255.255.255.252	PC↔FR
R-GB	S0/1/0	11.79.43.205	255.255.255.252	GB↔CI
R-CI	S0/0/0	11.79.43.206	255.255.255.252	GB↔CI
R-GB	S0/0/1	11.79.43.209	255.255.255.252	GB↔AR
R-AR	S0/0/0	11.79.43.210	255.255.255.252	GB↔AR
R-SP	Gi0/1	11.79.43.213	255.255.255.252	SP↔AR
R-AR	Gi0/1	11.79.43.214	255.255.255.252	SP↔AR

Servers (Photon Cove LAN)

Device	IP Address	Subnet Mask	Gateway	Role
SRV-WEB	11.79.43.10	255.255.255.128	11.79.43.1	Web Server
SRV-DNS	11.79.43.11	255.255.255.128	11.79.43.1	DNS Server
SRV-EMAL	11.79.43.12	255.255.255.128	11.79.43.1	Email Server
SRV-DHCP	11.79.43.13	255.255.255.128	11.79.43.1	DHCP Server

End Devices

Device	IP Address	Subnet Mask	Gateway	Method
PC-PC1	11.79.43.2	255.255.255.128	11.79.43.1	Static
PC-PC2	11.79.43.3	255.255.255.128	11.79.43.1	Static
PC-GB1	11.79.0.2	255.255.224.0	11.79.0.1	Static
PC-GB2	11.79.0.3	255.255.224.0	11.79.0.1	Static
PC-SP1	11.79.43.130	255.255.255.192	11.79.43.129	Static
PC-SP2	11.79.43.131	255.255.255.192	11.79.43.129	Static
PC-CI1	11.79.32.2	255.255.248.0	11.79.32.1	Static
PC-CI2	11.79.32.3	255.255.248.0	11.79.32.1	Static
PC-FR1	11.79.40.11+	255.255.254.0	11.79.40.1	DHCP
PC-FR2	11.79.40.12+	255.255.254.0	11.79.40.1	DHCP
PC-AR1	11.79.42.11+	255.255.255.0	11.79.42.1	DHCP
PC-AR2	11.79.42.12+	255.255.255.0	11.79.42.1	DHCP

DHCP Pools (SRV-DHCP)

Pool	Network	Mask	Gateway	Start IP	Max Users	DNS
FROSTFIELD	11.79.40.0	255.255.254.0	11.79.40.1	11.79.40.10	502	11.79.43.11
AURORA	11.79.42.0	255.255.255.0	11.79.42.1	11.79.42.11	245	11.79.43.11

Router Configuration Commands

R-PC (Photon Cove)

hostname R-PC

! DHCP Pools for remote LANs

ip dhcp excluded-address 11.79.40.1 11.79.40.10

ip dhcp excluded-address 11.79.42.1 11.79.42.10

ip dhcp pool FROSTFIELD

network 11.79.40.0 255.255.254.0

default-router 11.79.40.1

dns-server 11.79.43.11

ip dhcp pool AURORA

network 11.79.42.0 255.255.255.0

default-router 11.79.42.1

dns-server 11.79.43.11

! LAN Interface

interface GigabitEthernet0/0

ip address 11.79.43.1 255.255.255.128

no shutdown

! WAN to R-SP (Sunspire Palace)

interface Serial0/0/0

ip address 11.79.43.194 255.255.255.252

clock rate 64000

no shutdown

! WAN to R-GB (Granite Borough)

interface Serial0/0/1

ip address 11.79.43.197 255.255.255.252

clock rate 64000

no shutdown

! WAN to R-FR (Frostfield Ranch)

interface Serial0/1/0

ip address 11.79.43.201 255.255.255.252

clock rate 64000

no shutdown

! RIP v2 for dynamic routes

router rip

version 2

network 11.0.0.0

no auto-summary

! Static: PC ↔ SP (recursive)

ip route 11.79.43.128 255.255.255.192 11.79.43.193

! Static: PC ↔ GB (directly attached exit interface)

ip route 11.79.0.0 255.255.224.0 Serial0/0/1

R-GB (Granite Borough)

hostname R-GB

! LAN Interface

interface GigabitEthernet0/0

ip address 11.79.0.1 255.255.224.0

no shutdown

! WAN to R-PC (Photon Cove)

interface Serial0/0/0

ip address 11.79.43.198 255.255.255.252

no shutdown

! WAN to R-AR (Aurora Ridge)

interface Serial0/0/1

ip address 11.79.43.209 255.255.255.252

clock rate 64000

no shutdown

! WAN to R-CI (Corsair Isles)

interface Serial0/1/0

ip address 11.79.43.205 255.255.255.252

clock rate 64000

no shutdown

! RIP v2

router rip

version 2

network 11.0.0.0

no auto-summary

! Static: GB ↔ PC (directly attached exit interface)

ip route 11.79.43.0 255.255.255.128 Serial0/0/0

R-SP (Sunspire Palace)

hostname R-SP

! LAN Interface

interface GigabitEthernet0/0

ip address 11.79.43.129 255.255.255.192

no shutdown

! WAN to R-PC (Photon Cove) — Serial

interface Serial0/0/0

ip address 11.79.43.193 255.255.255.252

no shutdown

! WAN to R-AR (Aurora Ridge) — Gigabit

interface GigabitEthernet0/1

ip address 11.79.43.213 255.255.255.252

no shutdown

! RIP v2

router rip

version 2

network 11.0.0.0

no auto-summary

! Static: SP ↔ PC (recursive)

ip route 11.79.43.0 255.255.255.128 11.79.43.194

! Static: SP ↔ AR (recursive)

ip route 11.79.42.0 255.255.255.0 11.79.43.214

! Static: SP → GB Primary (recursive via PC)

ip route 11.79.0.0 255.255.224.0 11.79.43.194

! Static: SP → GB Floating (via AR, AD=200)

ip route 11.79.0.0 255.255.224.0 11.79.43.214 200

R-AR (Aurora Ridge)

hostname R-AR

! LAN Interface with DHCP helper

interface GigabitEthernet0/0

ip address 11.79.42.1 255.255.255.0

ip helper-address 11.79.43.1

ip helper-address 11.79.43.13

no shutdown

! WAN to R-SP (Sunspire Palace)

interface GigabitEthernet0/1

ip address 11.79.43.214 255.255.255.252

no shutdown

! WAN to R-GB (Granite Borough) — Serial

interface Serial0/0/0

ip address 11.79.43.210 255.255.255.252

no shutdown

! RIP v2

router rip

version 2

network 11.0.0.0

no auto-summary

! Static: AR ↔ SP (recursive)

ip route 11.79.43.128 255.255.255.192 11.79.43.213

R-FR (Frostfield Ranch)

hostname R-FR

! LAN Interface with DHCP helper

interface GigabitEthernet0/0

ip address 11.79.40.1 255.255.254.0

ip helper-address 11.79.43.1

ip helper-address 11.79.43.13

no shutdown

! WAN to R-PC (Photon Cove) — Serial

interface Serial0/0/0

ip address 11.79.43.202 255.255.255.252

no shutdown

! RIP v2 only (no static routes needed)

router rip

version 2

network 11.0.0.0

no auto-summary

R-CI (Corsair Isles)

hostname R-CI

! LAN Interface

interface GigabitEthernet0/0

ip address 11.79.32.1 255.255.248.0

no shutdown

! WAN to R-GB (Granite Borough) — Serial

interface Serial0/0/0

ip address 11.79.43.206 255.255.255.252

no shutdown

! RIP v2 only (no static routes needed)

router rip

version 2

network 11.0.0.0

no auto-summary

Assumptions:

1. Base network is `11.79.0.0/16` — derived from the last 4 digits of the last group member's student ID being `1179`, split into two parts: 11 and 79, giving network `11.79.0.0/16`.
2. Shortest path tree is rooted at Photon Cove using Dijkstra's algorithm on the given distance matrix. The resulting 5 inter-router links are: PC↔SP (280mi), PC↔GB (60mi), PC↔FR (190mi), GB↔CI (30mi), GB↔AR (110mi).
3. A direct physical link PC↔SP was kept even though the strict MST would prefer CI↔SP (250mi total vs 280mi direct). This was done to simplify the SP↔PC recursive static route requirement — a direct link makes next-hop resolution straightforward without relying on RIP convergence first.
4. An additional Gigabit link R-SP↔R-AR was added beyond the shortest path tree to satisfy two requirements simultaneously: (a) the SP↔AR recursive static route requirement, and (b) the SP→GB floating static route via Aurora Ridge.
5. The DHCP pools (FROSTFIELD and AURORA) are configured on both R-PC router and SRV-DHCP server. Both `ip helper-address 11.79.43.1` and `ip helper-address 11.79.43.13` are configured on R-FR and R-AR so that either server can respond to DHCP requests.
6. DHCP excluded address ranges `11.79.40.1–11.79.40.10` and `11.79.42.1–11.79.42.10` were reserved to prevent the DHCP server from assigning the gateway IP or any other reserved addresses to clients.
7. Corsair Isles and Frostfield Ranch routers have no static routes — all routing for these locations is handled entirely by RIP v2.

8. The SP→GB primary static route uses next-hop `11.79.43.194` (via R-PC). The floating static route uses next-hop `11.79.43.214` with administrative distance 200 (via R-AR), so it only activates if the primary path via R-PC goes down |
9. `no auto-summary` is configured on all routers because the subnets cross classful boundaries — for example, `11.79.40.0/23` and `11.79.43.0/25` both fall within the Class A `11.0.0.0/8` block, so auto-summary would incorrectly advertise only the /8 |
10. Email service is restricted to Sunspire Palace, Photon Cove, and Granite Borough as specified in the requirements. In Packet Tracer, each email client must be manually configured with the correct username, domain, and SMTP/POP3 server address pointing to SRV-EMAL (`11.79.43.12`) |
11. All WAN links use /30 subnets carved from the remaining space in `11.79.43.192/27`, providing exactly 2 usable host addresses per point-to-point link — no addresses are wasted |
12. The DNS server (SRV-DNS) is assumed to be the authoritative resolver for the domain `luminara.net`. All end devices must have their DNS server set to `11.79.43.11` to resolve `www.luminara.net` → `11.79.43.10` |